To Execute Large Pipelines, Launch Multiple Jobs, Measure Stability, Monitor Resources, Detect Failures, Analyze Performance.

To Document Architecture, Create Flow Diagrams, Explain Execution Paths, Describe Syscalls, Record Design Decisions, Review Documentation.

```
varnikakomali@VARNIKAs-MacBook-Air skill % nano skill_23.c
varnikakomali@VARNIKAs-MacBook-Air skill % gcc -Wall -Wextra skill_23.c -o skill_23 && ./skill_23
=== Part 1: Large Multi-Stage Pipeline Execution & Timing ===

   11970
Pipeline: seq 100000 | grep 5 | grep 0 | sort -n | wc -l
Execution Status: STABLE & COMPLETE
Pipeline Latency: 36.69 ms

=== Part 2: Concurrent Job Saturation & Resource Usage ===

Spawned 8 concurrent child tasks successfully.
All concurrent child tasks reaped.
Concurrent Execution Wall Time: 153.55 ms
Resource Metrics (getrusage child cumulative):
  User CPU Time:    0.056269 s
  System CPU Time:  0.018592 s
  Peak Resident RSS:2544 KB
Job Concurrency Health: PASS (Zero Zombies / Zero Leaks)

=== Part 3: POSIX Shell Subsystem Architecture Reference ===

Execution Flow Diagram:
  [User Input Line]
         |
         v
  [Tokenizer & Syntax Parser] ---> (Traps ERR_EMPTY_PIPE, ERR_MISSING_REDIRECT)
         |
         v
  [Pipeline Allocator] ----------> pipe() array configured [2 * (N-1) fds]
         |
         v
  [Process Dispatcher] ----------> fork() -> setpgid() leader election
         |                                |
         +-- Parent Shell                 +-- Child Subprocess
         |   - tcsetpgrp() [FG]           |   - dup2() pipes / redirection
         |   - waitpid() [WUNTRACED]      |   - signal() SIG_DFL
         |   - job_table_update()         |   - execvp()
         v
  [State Synchronizer] ----------> sigaction(SIGINT, SIGTSTP, SA_RESTART)

Primary System Call Specifications:
  * pipe()     : Allocates unidirectional data channels between stage pairs.
  * dup2()     : Replaces standard descriptors (0, 1, 2) atomically.
  * setpgid()  : Decouples shell session from background/foreground job groups.
  * tcsetpgrp(): Arbitrates controlling terminal hardware access.
  * waitpid()  : Harvests status codes non-blocking (WNOHANG) or traced.
  * getrusage(): Monitors kernel page faults and CPU utilisation counters.

varnikakomali@VARNIKAs-MacBook-Air skill %
```